

Taynam Ghaleb AlZamel

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Designer & Engineer | AI Product Design | Rapid Prototyping | Research & Facilitation | Riyadh

PROFESSIONAL SUMMARY

Software engineering graduate from Prince Sultan University (AI & Data Science track), working at the intersection of AI products, design, and institutional partnerships. Currently in a co-op role at Sarj AI in Riyadh, where I designed the QA function and shipped a sentiment-aware call analysis system into the product. Founder of Ruwad, a Damascus University partnership supporting Syrian students in global engineering competitions. Experienced in rapid prototyping with Cursor and AI tools, semi-structured user research, and translating between engineers, government partners, and corporate sponsors. Selected for 1st place at the PSU AI Hackathon under the Ministry of Health, and for direct partnership with CEER Motors and Riyadh Municipality on a senior project now incorporating as a startup.

WORK EXPERIENCE

Associate, AI Quality & Product Operations | Sarj AI (Co-op) Dec 2024 – Present

- Designed and proposed the company's release process (test environments, documented test cases, engineering sign-off) — adopted by the team and now used as onboarding material for new hires at a 17-person company.
- Identified a structural flaw in client KPI reports where out-of-scope callers were labeled as agent failures. Proposed a reframe and shipped a sentiment analysis + scope classification system that now feeds into client-facing reports. Sarj's product includes sentiment analysis because of this work.
- Built a label verification tool, test scenario automation, and QA traceability layer across the platform. Moved between QA, product, and AI engineering work as the team needed.
- Translated client KPI complaints into product requirements; collaborated with engineering and operations to ship the refined measurement architecture into production.

Programming Head | Industry 4.0 Lab, Prince Sultan University Jan 2022 – Present

- Led multidisciplinary student teams across embedded, perception, and autonomy projects over a 3.5-year tenure. Acted as the technical interface between subteams and lab leadership.
- Designed and delivered 3+ technical workshops (C/C++, embedded systems, sensor integration) to onboard new lab members and accelerate delivery timelines.
- Led development of an autonomous agriculture robot recognized by the Saudi Ministry of Agriculture at ICSDI 2024 — applied innovation with real-world stakeholder validation.
- Built lab's official React website and an NFC-based progress logging system, improving operational visibility and traceability across the lab's portfolio of projects.

SELECTED PROJECTS

RASD — Road Safety Detection Designed for Saudi Conditions | Senior project · Partners: CEER Motors, Riyadh Municipality · Now incorporating as a startup · Sept – Nov 2025

- Stakeholder engagement:** Pitched and led a senior project partnership with CEER Motors (Saudi national EV brand) and Riyadh Municipality. Ran 3–4 months of weekly review cycles with senior ADAS engineers and Municipality stakeholders.
- System design:** Designed a multi-signal perception system (camera + LiDAR fusion) tuned for Saudi road conditions where European-trained ADAS stacks fail — degraded lane markings, mixed surfaces, unregistered speed bumps.
- Reframing the problem:** A Municipality requirement (geolocate every detected speed bump for legality audit) reframed the project from driver-assist to infrastructure intelligence — turning a feature into a public-sector spend opportunity.
- Two-stakeholder output:** Designed the system to serve two stakeholders simultaneously: real-time driver alerts in the vehicle, and geo-tagged maintenance reports for the Municipality's repair prioritization.

- **Outcome:** Demonstrated to CEER and formally reviewed twice by the Municipality. Now incorporating as a startup based on the Municipality's identification of the work as a public-sector procurement opportunity.

Ruwad & EcoShaheen — Building a Pathway That Didn't Exist | *Founder · Partnership with Damascus University · Shell Eco-marathon Doha qualifier · 2025 – Present*

- **Founded solo:** Founded Ruwad to enable Syrian students to participate in global engineering competitions, with end-to-end support spanning institutional partnership, sponsorship, training, and competition logistics.
- **Institutional partnership:** Negotiated and documented a formal partnership with Damascus University, providing institutional legitimacy needed for sponsor engagement and competition entry. Established active 10–20 person student engineering team.
- **EcoShaheen flagship:** Led first flagship team EcoShaheen to qualification for Shell Eco-marathon Doha — competing against teams from countries with decades of established pathways into this competition.
- **Sponsorship engagement design:** Designed the sponsorship proposal around sponsor risk perception (Why Now, concrete ROI, post-competition deliverables) rather than the standard team-talent pitch — actively iterating with corporate sponsors now.
- **Why this exists:** Syrian students of my generation don't have the institutional pathways into global engineering competitions that students in better-resourced countries do. Ruwad is the early work of building that pathway, one team at a time.

Etma'en — Medication Safety for a Healthcare System That Can't Share Data | *PSU AI Hackathon, under Ministry of Health · Team of 4 · 1st Place · 2024*

- Steered the team toward an **Apple-Pay-style architecture** (patient data stays on device, hardware-tap conflict check at point of care) after recognizing cross-hospital data consolidation is institutionally impossible in Saudi Arabia.
- **Research design:** Shaped interview questions around testable design hypotheses; translated findings from 4 stakeholder interviews (doctors, pharmacist, patient) and 71 survey responses into a structured requirements set with documented informed consent.
- **Hardware prototype:** Built the embedded hardware layer (Arduino + RFID) — the clinician-side tap reader that made the architecture demonstrable. Integrated Gemini API as a **natural language chatbot interface** against data that never left the phone.
- **Constraint thinking:** Earlier version was over-engineered. Pulled design back to its simplest expression — one tap, one device, one binary result. Discipline of designing for regulated institutional environments turned out to be the discipline of removing things.
- **Outcome:** 1st place at PSU AI Hackathon under the Ministry of Health's umbrella, validating both the architectural concept and the user research approach.

PSU Eco Team — Real-Time Race Telemetry Dashboard | *Shell Eco-marathon · Built with Cursor and AI tools · 2025 – Present*

- **AI-built frontend:** Built the live race telemetry dashboard using Cursor and AI tooling, going from brief to working interface fast enough to iterate with the driver and pit crew before race day.
- **Real-time data pipeline:** Integrated MQTT, GPS, and joulemeter feeds with an ML model that surfaces driving cues (throttle, coast, brake points) in real time to the driver.
- **Competition impact:** Contributed to 1st Place in Efficiency, Middle East, at Shell Eco-marathon Lusail — the dashboard supported real-time driver decision-making and post-run strategy analysis.

SAJI — Freelance Design Engagement (Multi-Role Retail System) | *Freelance · Jul 2023 – Jan 2024*

- **Multi-role design system:** Designed digital surfaces across three user types — customer-facing storefront, B2B account flow, and internal employee portal — with role-based dashboards and a shared visual language across all three.
- **Tools and approach:** Worked in Figma across visual system, typography, and component patterns. Translated business requirements into screen flows and worked directly with the client on iteration.

EDUCATION

Bachelor of Software Engineering — AI & Data Science Track | Prince Sultan University, Riyadh

- Relevant Coursework: Machine Learning, Big Data, Human-Computer Interaction, Project Management, Requirements Engineering
- Leadership: President, International Students Club

CERTIFICATIONS & TRAINING

- **Self-Driving Cars Specialization** — University of Toronto (2025)
- **Modeling of Autonomous Systems** — University of Colorado Boulder (2025)
- **Modern Robotics: Mechanics, Planning and Control** — Northwestern University (2024)
- **Modern Robotics: Foundation in Robotics Motion** — Northwestern University (2024)

AWARDS & RECOGNITION

- **1st Place** — PSU AI Hackathon under the Saudi Ministry of Health (2024)
- **1st Place** — Shell Eco-marathon Middle East, Efficiency Category, Lusail (2025)
- **Best Solution** — Saudi Space Agency Case Study Competition (2025)
- **1st Place** — Sahem Social Entrepreneurship Hackathon, King Khalid Foundation
- **Best ML Project** — CCIS Project Expo, Prince Sultan University
- **Best Code & Strategy Award** — Sumo Robot Competition (2024 & 2025)

TECHNICAL SKILLS

Design: Figma, Excalidraw, Adobe Creative Suite (growing proficiency), visual systems, typography, component patterns, prototyping.

AI Prototyping: Cursor, Claude, v0, Lovable. Comfortable building end-to-end working prototypes from a brief.

Research & Facilitation: Semi-structured interviews, survey design, usability evaluation, requirements engineering, stakeholder workshops, working with government and industry partners.

Engineering: Python, C++, JavaScript, ROS2, embedded systems (Arduino, RFID), computer vision (YOLO), MQTT, Node-RED, real-time data pipelines, sensor fusion. Used when needed to ship a prototype rather than describe one.

Languages: Arabic (native), English (fluent).